

Hotel Bar Inventory Forecasting and Par Level Recommendations

Data period	01 Jan 2023 - 01 Jan 2024
Coverage	6 bars, 16 brands, 96 series
Inventory equation pass rate	100.0%
Final holdout WAPE	169.3%
Policy test stockout days	255
Policy test lost demand	42,874 ml
Average simulated closing inventory	467 ml

Business problem and approach

The goal is to reduce guest-facing stockouts without tying up cash in slow-moving bottles. Transaction records were validated against $\text{Closing} = \text{Opening} + \text{Purchase} - \text{Consumed}$, rolled into a complete daily bar-brand demand grid, and explicitly filled with zero-demand days. The historical data contains 999 close-to-zero closing-balance events; these are used as stockout proxies, not proof of lost sales.

Demand model

For each bar-brand series, two transparent causal models competed using time-ordered training data: a 7-day moving average and a rolling same-weekday mean. The lower-WAPE model was selected, then evaluated on the final 20% of days held out from selection. Overall test WAPE is 169.3%. Model selections: 7-day moving average: 59, day-of-week seasonal mean: 37. These simple models are stable with zeros and easy for managers to explain.

Par-level policy

The recommended par level is rounded to 30 ml and equals forecast $\times$ 2 delivery days plus safety stock: $1.645 \times \text{RMSE} \times \sqrt{2}$. This targets a 95% cycle-service level. Each day, managers order up to par based on inventory position, defined as on-hand plus confirmed inbound stock.

Backtest and operational interpretation

A daily-review simulation over the final 20% of days starts each series at its calculated par and receives orders after two days. It produces 255 stockout days, 42,874 ml of unmet observed demand, and 467 ml average closing inventory. This is a policy scenario, not a claimed historical improvement: recorded consumption may already be suppressed on stockout days.

Assumptions, risks, and next steps

Lead time is fixed at two days; unmet demand is lost; inventory is measured consistently in ml; and no promotions, holidays, supplier constraints, waste, or substitutions are observed. In production, refresh daily after POS close, capture on-hand and purchase-order data, alert on large forecast error or stockout proxies, and add event calendars and variable lead-time distributions. Refit or champion-challenge models monthly and monitor WAPE, service level, inventory days, and shrinkage.